

A Small Open-Economy DSGE Model for the UK

Working Paper No. 1 - Stage 1: System Estimation and Preliminary Forecasting

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1 1. Introduction

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2 2. The Theoretical Structural Model

Following the structural modeling tradition of @Scarth1988, we define the UK economy as a Small Open Economy (SOE) characterized by nominal rigidities and rational expectations. The model consists of a system of stochastic behavioural equations and non-stochastic national accounting identities.

2.1 2.1 Stochastic Behavioural Equations

The core dynamics are driven by forward-looking agents and the Mundell-Fleming extension for open-economy linkages. All variables (unless otherwise noted) are expressed as log-deviations from their steady-state trends.

The Open-Economy IS Curve: Aggregate demand depends on expected future output, real interest rates, and the real exchange rate (the expenditure-switching effect):

$$y_t = E_t y_{t+1} - \sigma(r_t - E_t \pi_{t+1}) + \gamma q_t + \epsilon_t^y$$

where q_t is the real exchange rate and σ represents the intertemporal elasticity of substitution.

The Rational Expectations Phillips Curve (REPC): Inflation is driven by forward-looking expectations and domestic demand pressure (the output gap):

$$\pi_t = \beta E_t \pi_{t+1} + \kappa y_t + \epsilon_t^\pi$$

Uncovered Interest Parity (UIP): To close the capital account under rational expectations, the domestic interest rate differential must equal the expected depreciation of the currency:

$$r_t - r_t^* = E_t e_{t+1} - e_t + \phi_t$$

where ϕ_t represents a stochastic risk premium.

2.2 2.2 National Accounting Identities

A critical departure in this paper is the explicit inclusion of national accounting identities as non-stochastic constraints within the 3SLS estimation framework. These identities ensure the structural model remains consistent with the ONS dataset.

The Production-Tax Identity: We define Gross Domestic Product (y_t) as the sum of the Gross Value Added of industries (y_t^d) and net taxes on products (T_t):

$$y_t = \omega y_t^d + (1 - \omega) T_t$$

where ω represents the steady-state share of GVA in total output.

The Expenditure Identity: To ensure the IS story is fully closed, aggregate output must also satisfy the expenditure equilibrium:

$$y_t = c_y c_t + i_y i_t + g_y g_t + n x_t$$

where the coefficients denote the shares of consumption, investment, government spending, and net exports in total GDP.

By defining both the production side ($y^d + T$) and the expenditure side ($c + i + g + nx$), the model accounts for the “double-entry” nature of the UK national accounts, providing a more robust set of instruments for the 3SLS estimation.

3 3. The empirical model

3.1 3.1 Variables

```
echo=FALSE
message=FALSE
warning=FALSE

library(knitr)
library(tibble)

var_mapping <- tribble(
  ~"Math Notation", ~"R Variable",      ~"Description",      ~"Classification",
  "$y_t$",          "Y_GAP",            "Output Gap",        "Endogenous",
  "$\\pi_t$",       "Infl_dev",        "Inflation Deviation", "Endogenous",
  "$\\Delta i_t$",  "d_BOE_rate",      "Change in Bank Rate", "Endogenous",
  "$u_t$",          "U_GAP",            "Unemployment Gap",   "Endogenous",
  "$r_t$",          "real_BOE_r",       "Ex-ante Real Interest Rate", "Exogenous",
  "$spr_t$",        "d_mkt_spr",        "Market Credit Spread", "Exogenous",
  "$\\Delta nx_t$", "dlog_NX",          "Growth in Net Exports", "Exogenous",
  "$q_t$",          "rer_gap",          "Real Exchange Rate Gap", "Exogenous",
  "$c_t, i_t, g_t$", "c_gap, i_g, g_g",  "Expenditure Gap Components", "Identity Constrained",
  "$y^d_t, T_t$",   "gva_g, tax_g",     "Production/Tax Components", "Identity Constrained",
  "$\\pi^e_t$",     "dlog_energy_cpi",  "Energy Price Inflation", "Instrument",
  "$\\Delta m_t$",  "dlog_IM_defl",     "Import Price Growth", "Instrument",
  "$\\Delta c_t$",  "dlog_cap",         "Capacity Utilization Growth", "Instrument",
  "$D_{2020}$",     "dummy_2020Q2/3",  "Structural Dummies (COVID)", "Exogenous"
)

kable(var_mapping, escape = FALSE, caption = "Table 1: System Variables: Mapping and Classification")
```

Table 1: System Variables: Mapping and Classification

Math Notation	R Variable	Description	Classification
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Math Notation	R Variable	Description	Classification
y_t	Y_GAP	Output Gap	Endogenous
π_t	Infl_dev	Inflation Deviation	Endogenous
Δi_t	d_BOE_rate	Change in Bank Rate	Endogenous
u_t	U_GAP	Unemployment Gap	Endogenous
r_t	real_BOE_r	Ex-ante Real Interest Rate	Exogenous
spr_t	d_mkt_spr	Market Credit Spread	Exogenous
Δnx_t	dlog_NX	Growth in Net Exports	Exogenous
q_t	rer_gap	Real Exchange Rate Gap	Exogenous
c_t, i_t, g_t	c_gap, i_g, g_g	Expenditure Gap Components	Identity Constrained
y_t^d, T_t	gva_g, tax_g	Production/Tax Components	Identity Constrained
π_t^e	dlog_energy_cpi	Energy Price Inflation	Instrument
Δm_t	dlog_IM_defl	Import Price Growth	Instrument
Δc_t	dlog_cap	Capacity Utilization Growth	Instrument
D_{2020}	dummy_2020Q2/3	Structural Dummies (COVID)	Exogenous

3.2 3.2 Structural Equations

The structural system for our model is defined as follows:

3.2.1 National Accounting Constraints

To ensure the empirical model is consistent with UK National Accounts, the estimated output gap (y_t) is subject to two fundamental identities:

$$y_t = \omega y_t^d + (1 - \omega)\tau_t + \xi_{p,t} \quad (\text{Production Constraint})$$

$$y_t = w_c c_t + w_i i_t + w_g g_t + w_{nx} nx_t + \xi_{e,t} \quad (\text{Expenditure Constraint})$$

Where ω and w_j are the steady-state shares derived from the ONS dataset, and ξ_t represents the statistical discrepancy (the 'check' variables in our data).

3.2.2 Behavioral Equations

The core dynamics are captured by the following four-equation system, which incorporates both forward-looking rational expectations and backward-looking persistence:

$$\text{IS Curve: } y_t = \beta_1 y_{t-1} + \beta_2 y_{t+1|t} + \beta_3 r_t + \beta_4 spr_t + \beta_5 \Delta nx_t + \beta_6 q_{t-1} + \mathbf{D}\gamma_{IS} + \epsilon_{y,t}$$

$$\text{Phillips Curve: } \pi_t = \delta_1 \pi_{t-1} + \delta_2 \pi_{t-2} + \delta_3 y_t + \delta_4 \pi_t^e + \delta_5 \Delta m_t + \delta_6 \Delta c_t + \epsilon_{\pi,t}$$

$$\text{Taylor Rule: } \Delta i_t = \theta_1 i_{t-1} + \theta_2 \pi_t + \theta_3 y_t + \theta_4 spr_t + \gamma_3 D_{20Q2} + \epsilon_{i,t}$$

$$\text{Okun's Law: } u_t = \alpha_1 u_{t-1} + \alpha_2 y_t + \epsilon_{u,t}$$

Where ω and w_j are the steady-state shares derived from the ONS dataset, and ξ_t represents the statistical discrepancy (the 'check' variables in our data).

3.3 3.3 Data

```
DSGE_3SLS_DATA <- read.csv("C:\\Users\\hogan\\OneDrive\\Documents\\1 OLLIE\\0. R\\UK_economic_
```

3.4 3.4 3SLS model estimation

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3.5 3.5 Review of robustness

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4 4. Simulations and impulse response functions

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5 5. Forecasts and forecasting performance

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6 6. Conclusion and next steps

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